



A reduced reaction mechanism of biodiesel surrogates with low temperature chemistry for multidimensional engine simulation

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ABSTRACT

A reduced biodiesel mechanism composed of 156 species and 589 reactions is reduced from an original complex mechanism (3299 species and 10806 reactions) based on MD, MD9D, and *n*-heptane as the surrogates. The mechanism reduction is conducted using the path flux analysis method, which considers multiple reaction path generations in the analysis of species interactions, and isomer lumping. Calculations of homogeneous auto-ignition and perfectly stirred reactor (PSR) combustion on a variety of reaction states, including pressures from 1 to 100 atm and equivalence ratios from 0.5 to 2, are the basis of the reduction. The initial temperatures are from 700 to 1800 K for the auto-ignition, and the inlet temperature is 300 K for the PSR. These reaction states cover the high-pressure and low-temperature operating conditions of future engines using advanced combustion technologies characterized by fuel–air premixing and auto-ignition. The fidelity of the resulting reduced mechanism with low-temperature chemistry is examined using a variety of applications. Close agreements between the reduced and original mechanisms are obtained in the predictions of ignition delay, history of mixture temperature, and species mole fraction during homogeneous auto-ignition and the temperature profile in PSR. The reduced mechanism, further integrated with a nitrogen oxides chemistry and a two-step soot model, is implemented into the KIVA/CHEMKIN program for the 3D simulation of biodiesel spray combustion. The predicted liquid and vapor penetrations agree with the experimental data in a non-reactive biodiesel spray simulation, indicating an accurate estimation of biodiesel physical properties. In the simulation of biodiesel spray combustion, predicted spatial distributions of hydroxyl radical and soot also agree with the corresponding experimental data.

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1. Introduction

To further enhance thermal efficiency and reduce emission levels, future international combustion engines tend to rely on advanced combustion technologies characterized by fuel–air premixing and volumetric auto-ignition under low-temperature and high-pressure operating conditions [1,2]. The combustion processes under such extreme conditions are strongly determined by fuel reactivity, which is in turn a function of fuel molecular structure. The adaptability to renewable alternative fuels will also be emphasized for future engines to enhance energy sustainability. Thus, fueling of renewable fuels in future engines can inevitably lead to previously unexplored turbulence–chemistry interactions and new turbulent flame regimes. To assist the design of future engines

using renewable fuels, such small-scale turbulence–chemistry interactions need to be accurately accounted for in multidimensional engine simulation, in addition to the in-cylinder dynamics of engine cylinder scales. Therefore, stringent requirement has been imposed on the accuracy of the combustion reaction mechanism of renewable fuels for engine simulation.

Biodiesel has been well recognized as a promising alternative fuel for conventional compression ignition (CI) engines without the need of major modification to engine design. The fuel is oxygenated and composed of numerous components with various carbon chain lengths and differs in composition determined by the raw material (e.g., vegetable oils and animal fats) for production. Despite slightly increased emission of nitrogen oxides (NO_x) at certain operating conditions [3,4], fueling of biodiesel (usually by mixing with diesel fuel) in CI engines can effectively reduce the emissions of unburned hydrocarbon, carbon monoxide, and particulate matters [5–7]. In addition, the fuel also contains no sulfur elements, producing no sulfur oxides or sulfides.

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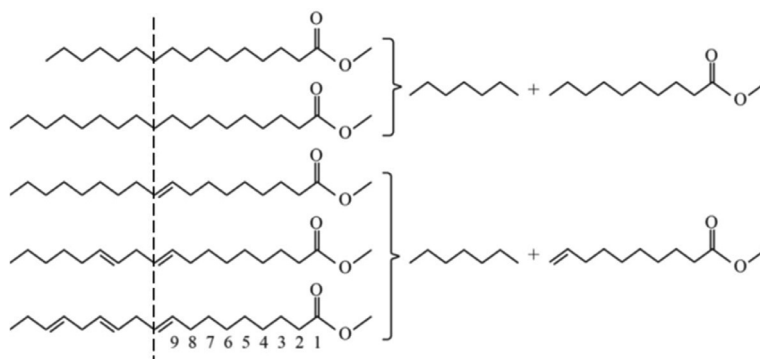


Fig. 1. Biodiesel combustion mechanism based on MD, MD9D, and *n*-heptane as the surrogates.

Biodiesel derived from soybean and rapeseed is mainly composed of two saturated fatty acid methyl esters, i.e., methyl palmitate ($C_{17}H_{34}O_2$) and methyl stearate ($C_{19}H_{38}O_2$), and three unsaturated fatty acid methyl esters, i.e., methyl oleate ($C_{19}H_{36}O_2$), methyl linoleate ($C_{19}H_{34}O_2$), and methyl linolenate ($C_{19}H_{32}O_2$), featuring long carbon chains with an ester group and different levels of unsaturation.

Considering the large molecular size of biodiesel methyl esters, a more practical way to simulate biodiesel combustion is to find appropriate surrogates and develop the chemistry based on the surrogates. Commonly used surrogates for biodiesel mechanisms are mainly from two categories of molecules, i.e., methyl esters and *n*-alkanes. Methyl butanoate (MB) has been used as the surrogate to simulate the biodiesel combustion, because of the existence of an ester group in its molecular structure [8–10]. However, limited by its short carbon chain length, MB has been found to be insufficient for biodiesel combustion modeling, primarily due to its lower reactivity compared to soybean biodiesel. *n*-hexadecane is the long-chain alkane found to be a good surrogate for rapeseed methyl ester (RME) [11,12]. The mechanism of *n*-hexadecane shows a reactivity in an agreement with that of actual RME measured by the experiment, but it failed to predict the early production of CO_2 due to the ester group, as observed in the same experiment. Consequently, according to the performance of the above two categories of surrogates, methyl esters with long carbon chain are the optimum surrogates for biodiesel.

In the molecular structure of some soy and rapeseed methyl esters, there can be multiple double bonds, leading to different levels of unsaturation. The existence of unsaturated fatty acid methyl esters in the fuel can have a strong negative impact on the overall reactivity of the fuel–air mixture, especially in the low temperature regime [13]. Double bonds can lead to the formation of unsaturated species which can also be the precursors of soot particles. Therefore, a biodiesel reaction mechanism needs to carefully deal with the effects of unsaturation. Biodiesel mechanisms based on the mixture of various types of surrogates have been proposed to consider the combined effects of long carbon chain, ester group, and unsaturation. Golovitchev and Yang [14] modeled RME as the mixture of methyl butanoate, *n*-heptane, and phenyl methyl ether (C_7H_8O). Ng et al. [15] proposed a compact biodiesel–diesel combined reaction mechanism based on MB, methyl crotonate, and *n*-heptane, which represent saturated fatty acid methyl ester, unsaturated fatty acid methyl ester, and straight-chain hydrocarbon, respectively. A complex mechanism including 3299 species and 10806 reactions was developed by Lawrence Livermore National Laboratory (LLNL) using the mixture of methyl decanoate (MD) as a saturated surrogate, methyl-9-decenoate (MD9D) as an unsaturated surrogate, and *n*-heptane as an alkane surrogate for biodiesel fuels [16]. It is assumed that each of the five main components undergoes a rapid decomposition into *n*-heptane and

MD (or MD9D), as described in Fig. 1. This complex mechanism can emulate the features of long carbon chain and ester group of actual biodiesel methyl esters, and the unsaturation due to double bond can be considered by MD9D. However, due to the large numbers of species and reactions, this mechanism is computationally expensive and cannot be used in three-dimensional simulation of engine in-cylinder dynamics. Thus, reducing this complex mechanism to an appropriate size without significantly impairing the computational fidelity is desired.

The reduction of reaction mechanism has been extensively studied with various categories of methods developed over the past decades. Skeletal reduction is one of the categories, and it finds and eliminates unimportant species and reactions to finally obtain a computationally efficient mechanism without losing the key features of the original mechanism. Starting from a target species, this method estimates the relation between the target species and each of other species and reactions by calculating a correlation coefficient. If the calculated correlation coefficient is less than a prespecified error tolerance, this species or reaction is found to be unimportant and can be eliminated from the mechanism. Either a species or a reaction can be used as the starting point of the reduction process. Commonly used methods include the principal component analysis [17,18], Jacobian analysis [19], sensitivity analysis [20,21], directed relation graph (DRG) [22,23], and its various derived versions, such as the DRG with error propagation [24,25]. Note that the DRG method is characterized by a linear reduction time, making it highly efficient to reduce extremely large mechanisms, and is usually used as the first step in the mechanism reduction. Lu et al. [26] extended the DRG method with expert knowledge (DRGX) by specifying a specific error tolerance for each target species. The resulting DRGX method can produce a skeletal mechanism capable of highly accurate predictions of heat release and concentrations of the target species, while moderate accuracies are also achieved for other species. Note that the DRG based skeletal method only considers the connection between the target species and species directly related. However, species not directly related to the target species can also have strong impacts on the target species via intermediate species, and it is valuable to consider this factor in the reduction process to further improve the accuracy of the reduced mechanism. Based on the second or even higher generations of reaction fluxes, Sun et al. [27] developed the path flux analysis (PFA) method to identify the importance of species indirectly related to the target species. Compared to the DRG method, improved accuracy of the reduced mechanism was achieved using the PFA method considering the first two generations of fluxes, without increasing the mechanism size. In addition, isomer lumping [28,29] is another category of reduction method, and it lumps similar species and reactions to reduce the number of variables that need to be tracked in chemical reaction computation.

The DRG-based methods have been used to develop reduced biodiesel reaction mechanisms for high-temperature conditions. Using DRG, isomer lumping, and optimized DRG-aided sensitivity analysis, Luo et al. [30] developed a reduced biodiesel mechanism including 118 species and 837 reactions from the original LLNL mechanism. The reduced biodiesel mechanism is only valid for temperatures above 1000 K. However, considering the dominance of low-temperature ignition in future engines using advanced combustion technologies, mechanisms only valid for high-temperature conditions are not applicable for the corresponding numerical studies. Under low-temperature (below 1000 K) and high-pressure conditions, determined by the long-chain molecular structures of major biodiesel components, elementary reactions can lead to strong negative temperature coefficient (NTC) effect, which is manifested by the short ignition delay comparable to the combustion time scale in engines. The low-temperature auto-ignition of biodiesel can have a strong impact on the engine combustion process, and thus the reduced biodiesel mechanism also needs to be valid for conditions below 1000 K. Since the low-temperature chemistry involves numerous key radicals and intermediate species, as well as complex reaction pathways, the reduced biodiesel reaction mechanism with low-temperature chemistry is significantly larger than those only valid for high-temperature conditions. Compact biodiesel schemes, including less than 100 species and 200 reactions, have been developed using the DRG method for multidimensional engine simulation [31–33]. The accuracy of these compact mechanisms was found to be insufficient, due to either the limited mechanism size or inappropriate biodiesel surrogates (e.g. MB). Using the DRG-based reduction methods, Luo et al. [34,35] obtained larger reduced biodiesel mechanisms (123 species and 394 reactions) with low-temperature chemistry. However, as mentioned above, the limitation of DRG-based methods is that only direct species interactions are analyzed. Considering the complexity of low-temperature chemistry, it is important to analyze indirect species interactions via intermediate species in the reduction of biodiesel mechanism.

The purpose of the present study is to develop a reduced biodiesel mechanism, which can be used in the numerical study of biodiesel applications in conventional CI engines and advanced engine combustion technologies characterized by volumetric auto-ignition and low-temperature combustion. The original biodiesel mechanism, including 3299 species and 10806 reactions, is the one developed by LLNL based on the three surrogates, i.e., MD, MD9D, and *n*-heptane, and it is reduced in the present study using the PFA method to consider multiple reaction path generations and isomer lumping. The resulting reduced mechanism is further combined with a NO_x chemistry and a two-step soot model for use in the multidimensional simulation of engine operation and emissions. The fidelity of the reduced mechanism is comprehensively examined by comparing with the original mechanism in calculating the combustion characteristics of various homogeneous applications. Experimental data of three-dimensional biodiesel spray flame under CI engine operating conditions are also used to validate the reduced mechanism.

2. Model formulation

2.1. Mechanism reduction

In the present study, the reduction of the biodiesel reaction mechanism is performed on a large range of reaction states, with pressures from 1 to 100 atm and equivalence ratios (Φ) from 0.5 to 2.0, representing lean, stoichiometric, and rich combustion conditions. The initial temperatures for homogeneous auto-ignition are from 700 to 1800 K, and the inlet temperature for the combustion in the perfectly stirred reactor (PSR) is 300 K. The large

range of reaction states ensures that the resulting reduced mechanism is applicable for the operating conditions of traditional CI engines and advanced engine combustion technologies characterized by volumetric auto-ignition and low-temperature combustion.

2.1.1. Path flux analysis reduction

The PFA method is used first to identify and remove relatively unimportant species and reactions from the original biodiesel reaction mechanism. As an alternative to the DRG approach, which is based on forward and reverse reaction rates, the PFA method uses production and consumption fluxes to evaluate the interaction between two species and identify important reaction pathways. The production flux, P_A , and consumption flux, C_A , for species A are defined by

$$P_A = \sum_{i=1}^I \max(0, \nu_{A,i} \omega_i) \quad (1)$$

and

$$C_A = \sum_{i=1}^I \max(0, -\nu_{A,i} \omega_i). \quad (2)$$

In the above two equations, I is the total number of reactions in which species A is involved. $\nu_{A,i}$ and ω_i are the stoichiometric coefficient and net production rate for species A by the i th reaction, respectively. The net production rate (ω_i) is the difference between forward ($\omega_{f,i}$) and reverse ($\omega_{b,i}$) reaction rates, i.e.,

$$\omega_i = \omega_{f,i} - \omega_{b,i}. \quad (3)$$

To evaluate the interaction coefficient, r_{AB} , which evaluates the error induced to the target species A due to the elimination of species B from the mechanism, the production and consumption fluxes of species A related to species B are calculated using the net production rate as

$$P_{AB} = \sum_{i=1}^I \max(0, \nu_{A,i} \omega_i \delta_{B,i}) \quad (4)$$

and

$$C_{AB} = \sum_{i=1}^I \max(0, -\nu_{A,i} \omega_i \delta_{B,i}). \quad (5)$$

$\delta_{B,i}$ is the delta function, which is equal to 1, if species A and B are simultaneously involved in the i th reaction, and 0, otherwise. Using the production and consumption fluxes, the interaction coefficients for the production and consumption of species A in direct relation with species B are defined by

$$r_{AB}^p = \frac{P_{AB}}{\max(P_A, C_A)} \quad (6)$$

and

$$r_{AB}^c = \frac{C_{AB}}{\max(P_A, C_A)}, \quad (7)$$

where superscripts “p” and “c” denote production and consumption, respectively.

The interaction coefficients determined by the above two equations only involve the first-generation reaction fluxes, which evaluate the directly relation with the target species A. However, species B can also be indirectly related to the target species A via intermediate species, involving the second or higher generations of reaction fluxes, and this indirect interaction can also be important to the production or consumption of species A. In the present study, in addition to the first-generation reaction flux, the second-generation reaction flux is also considered in the interaction analysis, and it has been proven that improved fidelity of the reduced

mechanism can be obtained by adding the second-generation reaction flux in the analysis [27]. Assume that species A and B are also indirectly related via a third species N_i , and the interaction coefficients for the second-generation are calculated by

$$r_{AB}^{p-2} = \sum_{N_i \neq A, B} (r_{AN_i}^p r_{N_iB}^p) \quad (8)$$

and

$$r_{AB}^{c-2} = \sum_{N_i \neq A, B} (r_{AN_i}^c r_{N_iB}^c). \quad (9)$$

The above interaction coefficients can be then used in the mechanism reduction with an error tolerance prespecified. Species B is found to be important to species A, if the interaction coefficient is greater than the error tolerance, and it is retained in the mechanism. Otherwise, the species is eliminated. The four interaction coefficients defined by the above equations are grouped into an overall interaction coefficient for species B, and thus only one error tolerance is needed, i.e.,

$$r_{AB} = r_{AB}^p + r_{AB}^c + r_{AB}^{p-2} + r_{AB}^{c-2}. \quad (10)$$

Since the purpose of the present study is to obtain a reduced biodiesel mechanism for use in the three-dimensional simulation of engine in-cylinder dynamics, it is important for the reduced mechanism to have as few species as possible to enhance the computational efficiency. On the other hand, considering the complexity of low-temperature chemistry, the reduced mechanism needs to be as detailed as possible to adequately capture the auto-ignition behavior under low-temperature conditions. Thus, reduction of the biodiesel reaction mechanism in the present study is performed to achieve a balance between computational accuracy and efficiency.

The target species for the PFA reduction in this study include the fuel, air (N_2 and O_2), CO, HO_2 , and C_2H_2 to accurately deal with the processes of fuel decomposition, oxidation, CO emission, H_2 – O_2 chemical reactions, and soot emission, respectively. Molar fractions of the three biodiesel surrogates, i.e., MD, MD9D, and n -heptane, in the mixture are prespecified to be 25%, 25%, and 50%, respectively. Despite the fact that the fuel surrogate composition for reduction does not significantly change the resulting mechanism, which also usually features good extensibility in predicting the ignition characteristics of fuels with different initial surrogate compositions [30,34], this surrogate composition for the present biodiesel mechanism reduction is selected to be intermediate, giving equal moles of saturated and unsaturated long-chain methyl esters according to the scheme in Fig. 1. A large threshold value of 45% is selected as the worst-case error tolerance for the PFA reduction to obtain a relatively compact reaction mechanism, and the resulting mechanism is composed of 185 species and 735 reactions after this reduction stage.

2.1.2. Isomer lumping

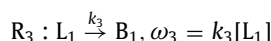
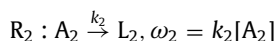
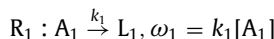
A further reduction of the mechanism is to consider the fact that the reaction mechanism of large hydrocarbons usually has many isomers, which have the same molecular weight and the same function groups. Since isomers also have similar transport properties, they can be grouped to obey a single transport equation. By grouping a few isomer species into one lumped species and eliminating the corresponding unimportant reactions, the number of variables to be tracked in combustion simulation can be effectively reduced. In this study, a simplified isomer lumping strategy is used to identify isomer groups, in which species have similar transport properties and participate in similar reactions [36]. Assume that two isomer species, for example, L_1 and

Table 1

Lumped isomers for the reduction of biodiesel reaction mechanism.

Lumped species	Isomer species
md3j	md3j, md4j, md5j, md6j, md7j, md8j, md9j
md3o2	md3o2, md5o2, md6o2
md3ooh5j	md3ooh5j, md5ooh3j, md6ooh4j
md3ooh5o2	md3ooh5o2, md5ooh3o2, md6ooh4o2
md9d3j	md9d3j, md9d4j, md9d6j, md9d7j
md9d3o2	md9d3o2, md9d6o2, md9d8o2, md8dxo2
c7h15-2	c7h15-2, c7h15-3, c7h15-4
c7h15o2-2	c7h15o2-2, c7h15o2-3
c7h14ooh2-4o2	c7h14ooh2-4o2, c7h14ooh3-5o2
nc7ket24	nc7ket24, nc7ket35
md9doh9	md9doh9, md9doh10

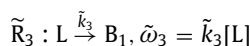
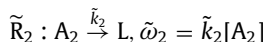
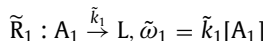
L_2 , participate in the following elementary reactions with the corresponding reaction rates, ω_i , i.e.,



k_i is the reaction rate coefficient of the i th reaction. If L_1 and L_2 can be lumped into species L , namely,

$$[L] = [L_1] + [L_2], \quad (12)$$

the above four reactions can be rewritten as



In the above equation, rate coefficients of the production reactions of isomers are simply $\tilde{k}_1 = k_1$ and $\tilde{k}_2 = k_2$, indicating that the lumping operation does not affect the production reaction of initial isomer species. Rate coefficients for the consumption reactions of isomers are adjusted by multiplying a factor proportional to the contribution of the isomer species in the lumped species composition, i.e., the ratio of the isomer species composition to that of the lumped species:

$$\tilde{k}_3 = k_3 \frac{[L_1]}{[L]}, \tilde{k}_4 = k_4 \frac{[L_2]}{[L]}. \quad (14)$$

Table 1 shows the isomer groups to be lumped in the biodiesel mechanism. The isomers are lumped only if the deviation of ignition delay from the original mechanism is less than 5% at all the validation conditions. After this reduction stage of isomer lumping, the resulting mechanism consists of 156 species and 589 reactions. To evaluate the production rate of each isomer species during the reaction computation, its concentration is needed and can be constructed from the lumped species concentration. The reconstruction is also a complex and unknown function of temperature, pressure, and concentrations of all other species. In this study, the reconstruction is approximated using a simple strategy according to Eq. (14), which assumes each isomer species concentration to be the product of a constant factor and the lumped species concentration [37].

Furthermore, to simulate the emission of NO_x during biodiesel combustion, a chemistry describing NO_x formation needs to be

present in the reaction mechanism. Since nitrogen is not contained in the fuel, a simple chemistry including 4 species and 12 reactions related to NO_x is integrated with the mechanism to model the thermal formation of NO_x [38]. Finally, the final reduced mechanism for biodiesel combustion is composed of 160 species and 601 reactions.

3. Results and discussion

3.1. Mechanism validation based on homogeneous applications

The present reduced biodiesel mechanism including 156 species and 589 reactions is first validated by comparing with the original complex mechanism in the calculations of homogeneous auto-ignition and PSR combustion at various combustion conditions. Figure 2 shows the predicted ignition delay as a function of initial temperature in a closed homogeneous reactor. The ignition delay is defined as the time when the temperature of the mixture increases by 400 K. The data are obtained at three pressures, representing low-, moderate-, and high-pressure conditions, and three equivalence ratios, representing lean, stoichiometric, and rich combustion conditions. Ignition delay curves predicted using the original mechanism at the same operating conditions are also shown in the same figure for the validation. The comparison indicates that the ignition delay data predicted by the reduced mechanism agrees closely with those by the original mechanism at all the selected conditions, especially in the low-temperature (below 1000 K) region which behaves strong NTC effect at high pressures. This close agreement proves that the present reduced mechanism can adequately reproduce the behavior of the original mechanism in predicting the low-temperature auto-ignition of biodiesel. Noted that no adjustment, which is frequently used to improve the prediction capability of reduced mechanisms, is made to the constants of the Arrhenius reaction rate coefficients. To further examine the fidelity of the reduced mechanism, predicted instantaneous mixture temperatures during the auto-ignition process from four different initial temperatures are also compared as shown in Fig. 3. The predicted results are for a pressure of 1 atm and an equivalent ratio of 1. It is seen that temperature histories predicted by the reduced mechanism also agree closely with those by the original mechanism for all the four initial temperatures.

Figure 4 shows the evolution of instantaneous molar fraction of a few important species during the auto-ignition of stoichiometric biodiesel-air mixture in the closed homogeneous reactor at constant pressure. Results are predicted using both the reduced and original mechanisms with equal molar fractions of MD, MD9D, and *n*-heptane as the fuel surrogate composition, different from the original composition used for mechanism reduction. The initial temperature of the reactor is 1600 K, and the pressure is fixed to be 1 atm. Comparison of the results by both mechanisms indicates that the reduced mechanism agrees with the original mechanism in predicting the mole fractions of all the selected important species under the current operating condition. This agreement also proves the extensibility of the present reduced mechanism to different compositions of fuel surrogate mixtures.

In addition to ignition, extinction is also a critical transient phenomenon for premixed combustion, and prediction of extinction is an important quality of a reaction mechanism. In the present study, the combustion in PSR at various operating conditions is simulated for a further validation of the present reduced mechanism. Figure 5 shows the comparison of calculated temperature profiles as a function of residence time in PSR at different equivalence ratios and pressures. For each of the selected conditions, the reduced mechanism shows a close agreement with the original complex mechanism in the calculation of the temperature profile, especially in the branch above the curve turning point,

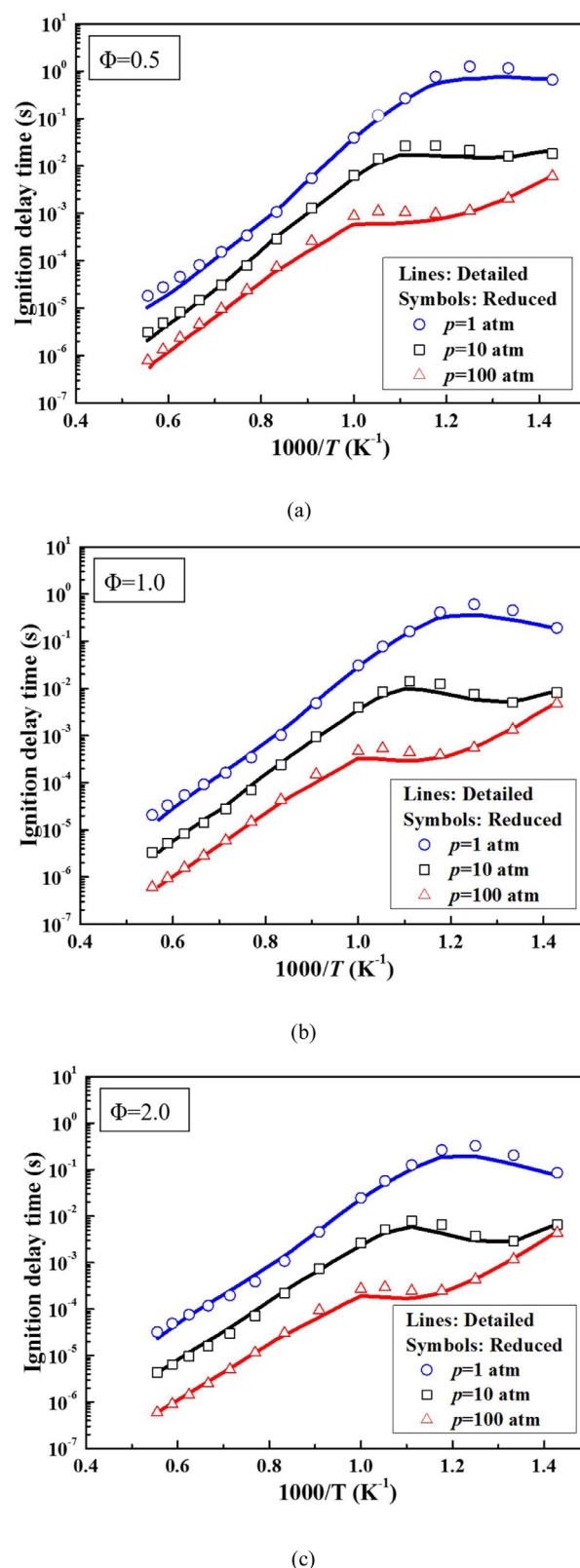


Fig. 2. Predicted ignition delay for biodiesel-air mixture using both the reduced and original mechanisms at pressures from 1 to 100 atm and equivalence ratios from 0.5 to 2.

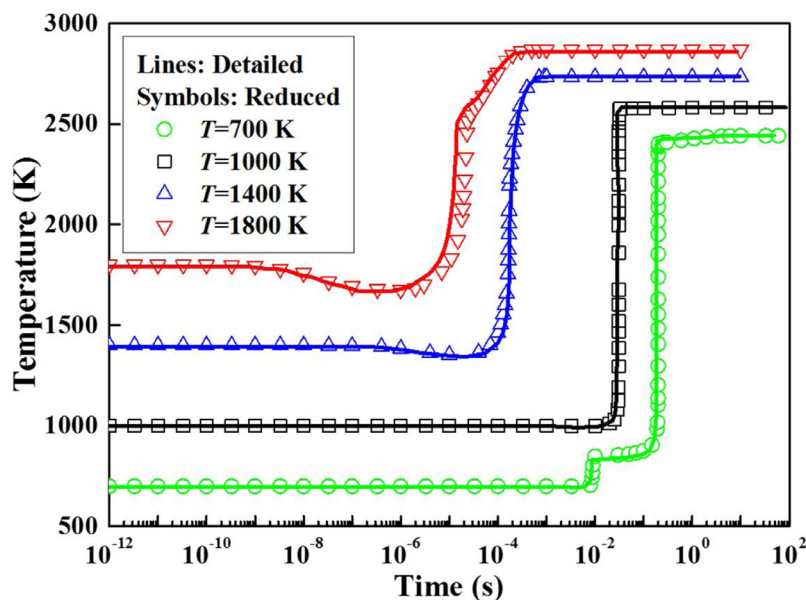


Fig. 3. Comparison of the predicted temperature histories from four different initial temperatures during auto-ignition by the reduced and original mechanisms.

Table 2
Operational parameters of the biodiesel spray combustion.

Fuel	SME
Injection orifice diameter	90.8 μm
Injection pressure	150 MPa
Volumetric compositions of ambient gas	Inert: $\text{O}_2=0\%$, $\text{N}_2=0.8971\%$, $\text{CO}_2=0.0652\%$, $\text{H}_2\text{O}=0.0377\%$ Reactive: $\text{O}_2=0.15\%$, $\text{N}_2=0.7515\%$, $\text{CO}_2=0.0622\%$, $\text{H}_2\text{O}=0.0363\%$
Gas density	22.8 kg m^{-3}
Gas temperature	900 K, 1000 K
Pressure	6.0 MPa, 6.7 MPa
Liquid fuel temperature	363 K
Injection fuel mass	22.7 mg
Discharge coefficient	0.94

which indicates the extinction of flame. This branch of the curve is a physical stable branch related to strong flames in realistic stable combustion systems operating in steady state. Noticeable deviation exists in the branch below the turning point for some operating conditions. Since this lower branch behaves simultaneous decreasing temperature and increasing residence time (or Damköhler number) from the extinction state, it is composed of physically unrealistic states and cannot be obtained experimentally. Thus, deviation in this branch does not impair the validity of the present reduced mechanism in the simulation of PSR combustion.

3.2. Mechanism validation based on three-dimensional biodiesel spray combustion

The present reduced biodiesel mechanism is also used to simulate a three-dimensional biodiesel spray combustion. The operating conditions of the simulation are consistent with those of the optically accessible experiment of fuel spray combustion conducted by Nerva et al. [39] in a constant-volume combustion chamber. The chamber has a cubical geometry, with an identical length of 108 mm in each dimension. Soy methyl ester (SME) biodiesel is injected into the chamber through an injector mounted on the top center of the chamber. The injector has a single hole with a diameter of 90 μm , and it is connected to a common rail, which fixes the injection pressure to be 150 MPa. The main operational parameters of the experiment are listed in Table 2.

Table 3
Composition of SME biodiesel used in the experiment [39].

Component	Mole fraction
Methyl Palmitate ($\text{C}_{17}\text{H}_{34}\text{O}_2$)	11.0%
Methyl Stearate ($\text{C}_{19}\text{H}_{38}\text{O}_2$)	4.0%
Methyl Oleate ($\text{C}_{19}\text{H}_{36}\text{O}_2$)	25.0%
Methyl Linoleate ($\text{C}_{19}\text{H}_{34}\text{O}_2$)	53.0%
Methyl Linolenate ($\text{C}_{19}\text{H}_{32}\text{O}_2$)	7.0%

The simulation is performed using an improved version of the KIVA-3V R2 code, with a CHEMKIN solver embedded in the program [40]. The present reduced biodiesel mechanism is then put in the KIVA/CHEMKIN program for the simulation of fuel spray combustion. The computational domain is simplified to be a cylindrical geometry with a diameter of 10 cm and a height of 10 cm, and the mesh is generated to have an average grid size of 1 mm. The gas-phase flow field is resolved using the Reynolds-Averaged Navier–Stokes equation, with the RNG $k-\varepsilon$ model for turbulence modeling. The breakup of biodiesel spray is simulated using the KH-RT model, which describes the breakup of liquid fuel spray into a two-stage process [41]. The soot emission is predicted using a two-step soot model [40], considering the generation and oxidation of soot particles. According to the scheme shown in Fig. 1, mole fractions of the three biodiesel surrogates in the combustion simulation are determined by the composition of the actual SME biodiesel used in the experiment as shown in Table 3. The

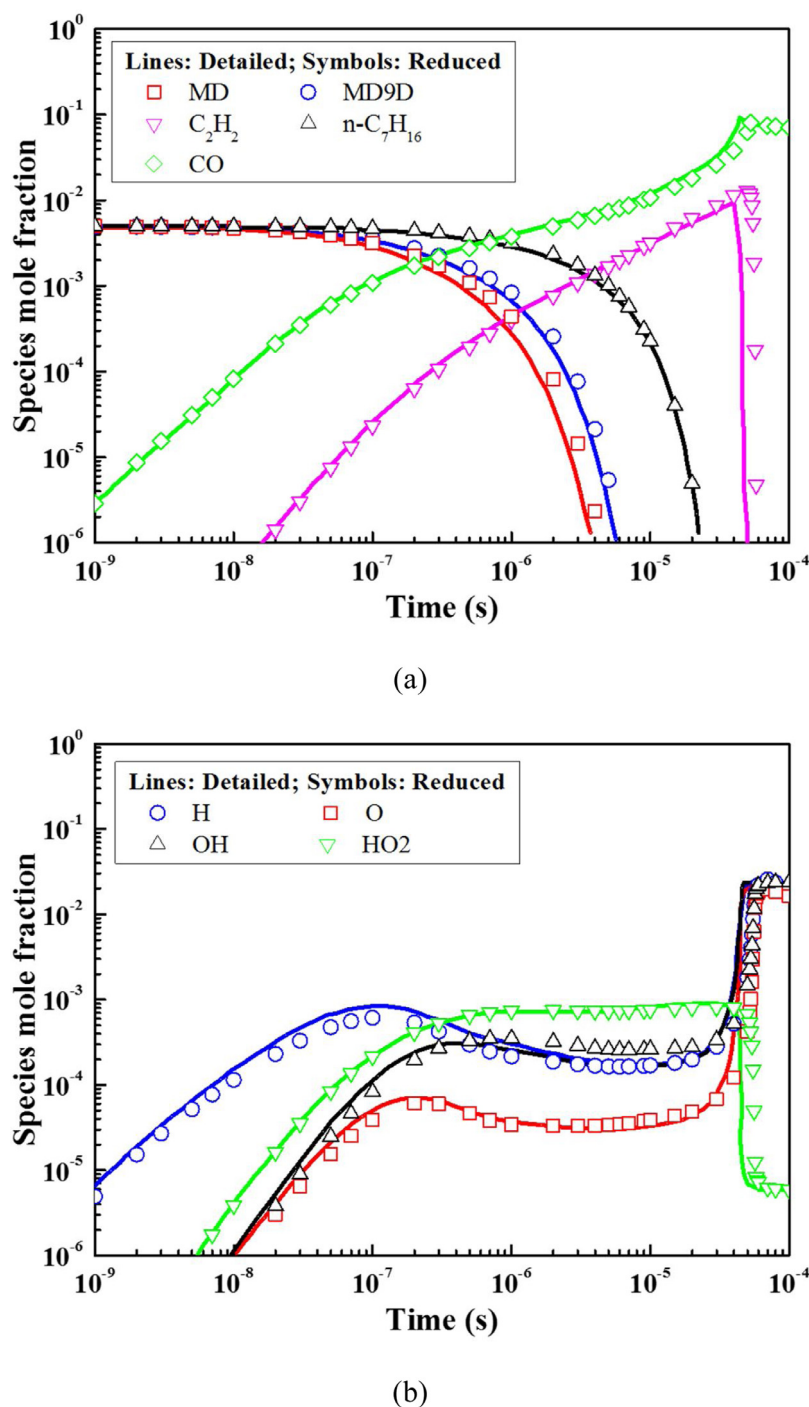


Fig. 4. Comparison of the molar fractions of a few important species during the constant-pressure auto-ignition from an initial temperature of 1600 K.

determination of surrogate mole fractions is to obey a simple rule that the carbon content and the ester group content are in balance with those of the actual biodiesel components [16]. The physical properties of biodiesel used in the spray modeling are calculated using the methods proposed by Zhang and Kong [42].

To verify first the calculation of biodiesel physical properties, the present models for biodiesel spray combustion are used to simulate a non-reactive biodiesel spray in an inert (no oxygen) ambience as shown in Table 2. Figure 6 shows the development of liquid and vapor penetrations of the spray at two different ambient temperatures. Also shown in the figure are the corresponding experimental data [39] for comparisons. The liquid penetration of

fuel spray is defined as the axial distance from the injector to the position behind which 95% of the total injected fuel exists, and the vapor penetration of fuel spray is the maximum axial distance from the injector to the position where the mass fraction of fuel vapor is greater than 0.001%. Spray penetration is a critical parameter evaluating the combustion performance: Excessive penetration may cause impingement of spray on cylinder wall, whereas deficient penetration can lead to insufficient atomization and mixing with ambient air. It is shown in Fig. 6 that liquid and vapor penetrations coincide at a very early stage of the biodiesel spray and start to separate shortly due to the vaporization of biodiesel drops. The liquid penetration fluctuates near a mean value, which is

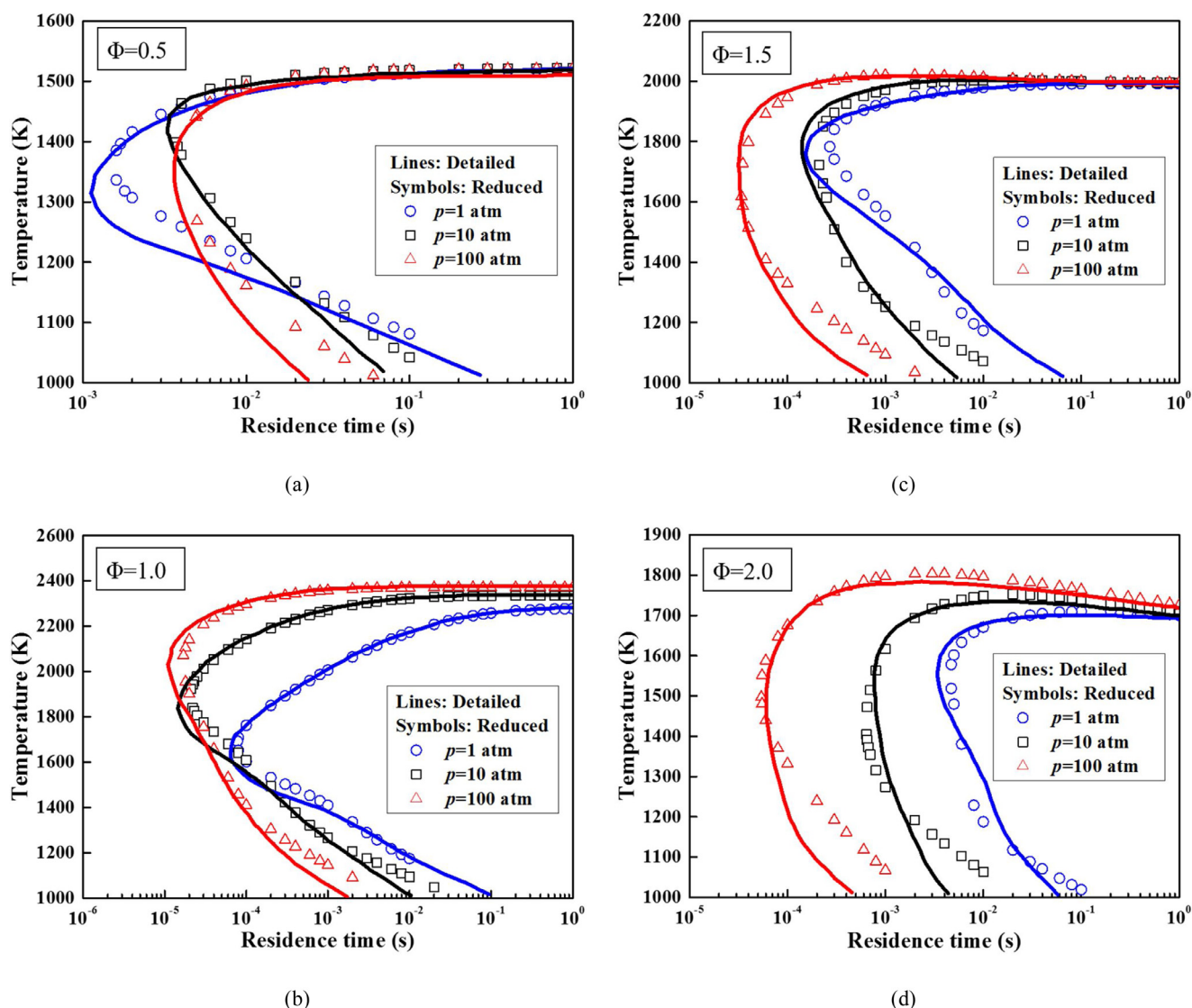


Fig. 5. Temperature profiles of biodiesel-air mixture combustion in PSR as a function of residence time for a fixed inlet temperature of 300 K and various pressures and equivalent ratios.

usually referred to as the liquid length. Beyond this axial distance, no liquid fuel drops exist in the cylinder, whereas the gas-phase fuel vapor continues to penetrate and mix with ambient air. The calculated liquid and vapor penetrations of the biodiesel spray at the two ambient temperatures are in good agreements with the experimental data, indicating accurate predictions of biodiesel physical properties.

Furthermore, the combustion of biodiesel spray in the reactive ambience (including 15% oxygen) is simulated by solving the present reduced mechanism using the CHEMKIN solver. Comparisons of predicted ignition delay and flame lift-off length with the corresponding experimental data at two ambient gas temperatures are shown in Table 4. Both predicted and experimental data show that higher ambient gas temperature leads to shorter ignition delay and flame lift-off length, indicating enhanced fuel spray atomization and ignition due to the higher ambient temperature. The predicted ignition delay at both ambient temperatures agrees with the experimental data, with the maximum relative error (RE) less than 10%. For both ambient temperatures, the present combustion model slightly overpredicts the flame lift-off length.

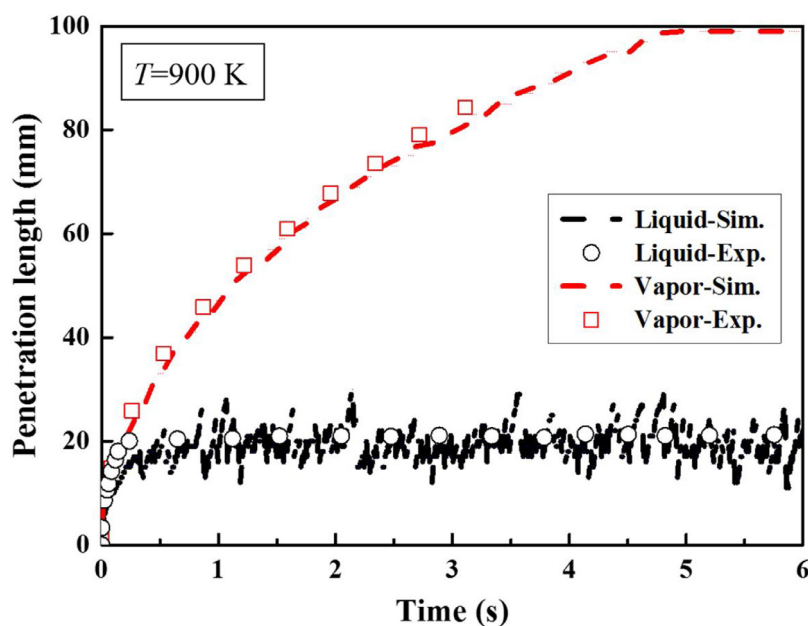
The flame lift-off lengths at both 900 and 1000 K ambient temperatures are also marked by the red dotted lines in Fig. 7, which

Table 4

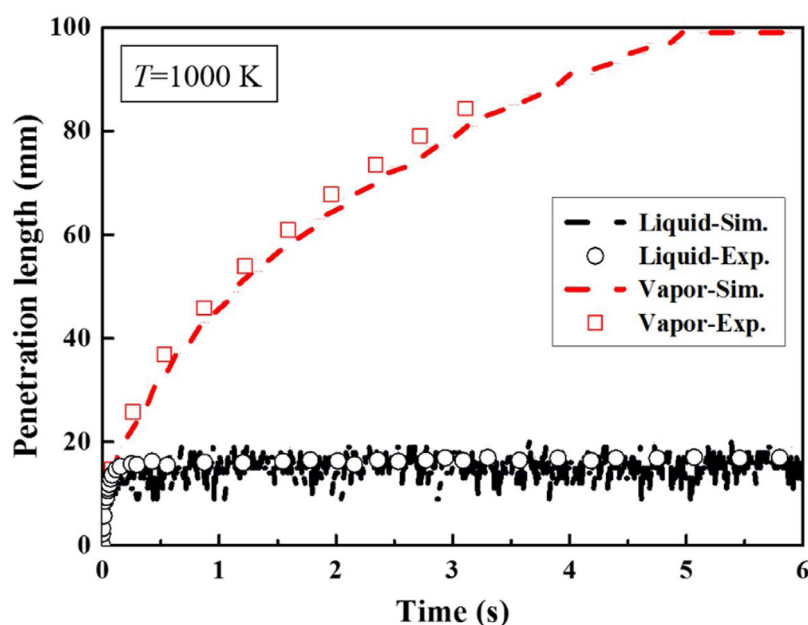
Comparison of the predicted ignition delay and flame lift-off length with experimental data at two initial temperatures.

Temp. (K)	Parameter	Exp.	Sim.	RE
900	Ignition delay (ms)	0.709	0.656	−7.48
	Flame lift-off length (mm)	26.18	28.58	9.17
1000	Ignition delay (ms)	0.377	0.384	1.86
	Flame lift-off length (mm)	17.27	21.91	26.87

shows both experimental and simulated images of hydroxyl radical (OH) distributions on the central cut plane of the cylindrical chamber at 3 ms after the start of injection (ASI). Note that quantities used to indicate the OH distributions in the experimental and simulated images are not identical. The experimental images are the ensemble average of OH chemiluminescence images proportional to the OH mass concentration, which is the quantity plotted in simulated images. Thus, quantities in the experimental and simulated images are equivalent in indicating the instantaneous spatial OH distribution. As shown by both experimental and simulated images, the OH concentrates in thin regions along the periphery of the flame, meaning high reaction rates in these regions. At the selected instant, the OH distributions predicted using the biodiesel



(a)



(b)

Fig. 6. Histories of liquid and vapor penetrations of biodiesel spray in comparison with experimental data at two ambient temperatures.

combustion models with the present reduced mechanism approximately agree with those shown by the experimental images.

In addition to the OH radical, experimental and simulated distributions of soot particles on the central cut plane of the cylinder at 3 ms ASI are also plotted for comparisons as shown in Fig. 8. The experimental images plot the contours of soot volume fraction measured using the planer laser-induced incandescence, whereas in the simulated images, the distribution of soot particles is indicated by the mass concentration. The soot emission is simulated using the two-step model with acetylene (C_2H_2) selected as the precursor species for the generation of soot particles. Soot distributions shown in both experimental and simulated images indicate that soot particles concentrate in the central fuel-rich region of the

biodiesel spray flame. The simulated soot distributions agree with those captured by the experiment in the overall geometry and axial position of the soot region and the relative intensity of concentration within the soot region.

4. Summary

In this study, a reduced biodiesel mechanism composed of 156 species and 589 reactions based on the mixture of three surrogates (MD, MD9D, and *n*-heptane) is developed by reducing the LLNL complex mechanism using the PFA reduction and isomer lumping. The selected three surrogates can adequately emulate the molecular structures of the five major biodiesel components

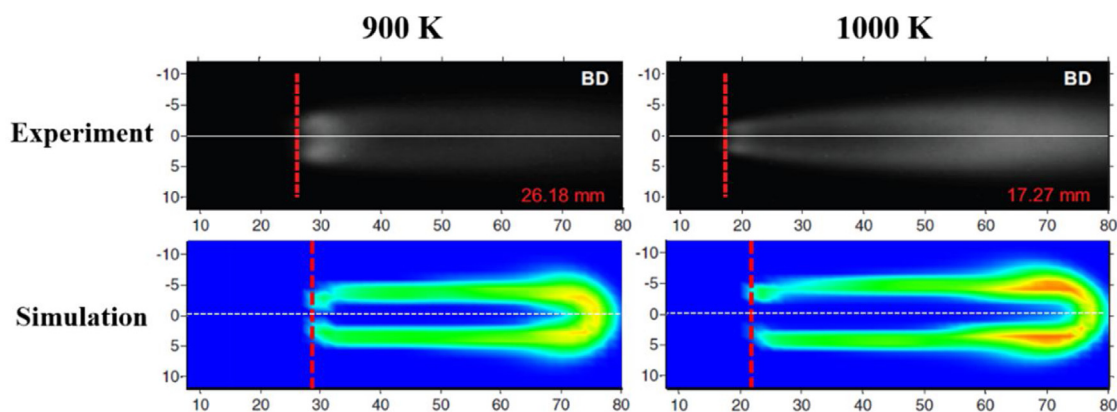


Fig. 7. Experimental and simulated distributions of OH at 3 ms ASI. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

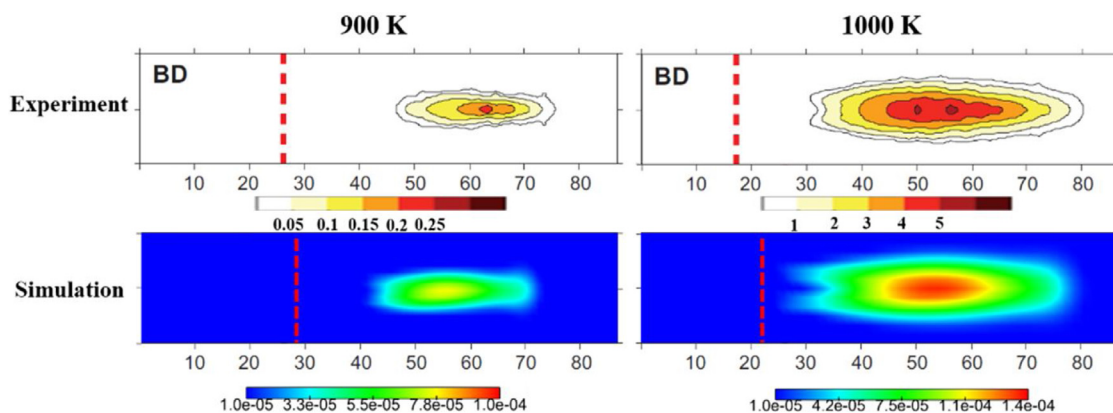


Fig. 8. Experimental and simulated distributions of soot particles at 3 ms ASI.

with different levels of unsaturation. Reduction of the mechanism is performed based on homogeneous auto-ignition and PSR combustion on a variety of combustion states covering a large range of operation conditions, especially in the region of low-temperature auto-ignition. The fidelity of the present reduced mechanism is first examined by comparing with the original complex mechanism in the simulations of homogeneous auto-ignition and PSR combustion under a variety of reaction conditions. Good agreements are obtained in the predicted ignition delay, history of mixture temperature, and history of major species concentration during auto-ignition. In the simulation of PSR combustion, close agreement is also obtained in the upper stable branch of the temperature profile. The present reduced mechanism, further integrated with a simple NO_x chemistry and a two-step soot model, is implemented into the KIVA/CHEMKIN program to simulate three-dimensional biodiesel spray combustion and emission under CI engine operating conditions. Agreements with experimental data are obtained in the liquid and vapor penetration lengths, flame lift-off lengths, and the spatial distributions of OH and soot. Consequently, the applicability of the present reduced biodiesel mechanism in the numerical study of biodiesel combustion and emission in conventional CI engine is demonstrated. Considering its accuracy in the prediction of low-temperature auto-ignition, the present reduced mechanism also has a great potential to be used in the future study of engines operating on advanced combustion technologies.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no signifi-

cant financial support for this work that could have influenced its outcome.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2019.11.002](https://doi.org/10.1016/j.combustflame.2019.11.002).

References

- [1] J. Chen, Petascale direct numerical simulation of turbulent combustion—fundamental insights towards predictive models, *Proc. Combust. Inst.* 33 (2011) 99–123.
- [2] R.D. Reitz, Directions in internal combustion engine research, *Combust. Flame* 160 (2013) 1–8.
- [3] W. Yuan, A.C. Hansen, Z. Tan, Modeling of NO_x emissions of biodiesel fuels, 2005 ASAE Annual International Meeting (2005) paper 056116.
- [4] J. Sun, J.A. Caton, T.J. Jacobs, Oxides of nitrogen emissions from biodiesel-fuelled diesel engines, *Prog. Energy Combust.* 36 (2010) 677–695.
- [5] E. Rajasekar, A. Murugesan, R. Subramanian, N. Nedunchezian, Review of NO_x reduction technologies in CI engines fuelled with oxygenated biomass fuels, *Renew. Sust. Energy Rev.* 14 (2010) 2113–2121.
- [6] A.K. Agarwal, Biofuels (alcohols and biodiesel) applications as fuels for internal combustion engines, *Prog. Energy Combust.* 33 (2007) 233–271.
- [7] C. He, Y. Ge, J. Tan, K. You, X. Han, J. Wang, Characteristics of polycyclic aromatic hydrocarbons emissions of diesel engine fueled with biodiesel and diesel, *Fuel* 89 (2010) 2040–2046.

- [8] E.M. Fisher, W.J. Pitz, H.J. Curran, C.K. Westbrook, Detailed chemical kinetic mechanisms for combustion of oxygenated fuels, *Proc. Combust. Inst.* 28 (2000) 1579–1586.
- [9] S. Gaïl, M.J. Thomson, S.M. Sarathy, S.A. Syed, P. Dagaut, P. Diévert, A.J. Marchese, F.L. Dryer, A wide-ranging kinetic modeling study of methyl butanoate combustion, *Proc. Combust. Inst.* 31 (2007) 305–311.
- [10] A. Farooq, D.F. Davidson, R.K. Hanson, L.K. Huynh, A. Violi, An experimental and computational study of methyl ester decomposition pathways using shock tubes, *Proc. Combust. Inst.* 32 (2009) 247–253.
- [11] P. Dagaut, S. Gaïl, Chemical kinetic study of the effect of a biofuel additive on Jet-A1 combustion, *J. Phys. Chem. A* 111 (2007) 3992–4000.
- [12] P. Dagaut, S. Gaïl, M. Sahasrabudhe, Rapeseed oil methyl ester oxidation over extended ranges of pressure, temperature, and equivalence ratio: experimental and modeling kinetic study, *Proc. Combust. Inst.* 31 (2007) 2955–2961.
- [13] Y. Zhang, Y. Yang, A.L. Boehman, Premixed ignition behavior of C₉ fatty acid esters: a motored engine study, *Combust. Flame* 156 (2009) 1202–1213.
- [14] V.I. Golovitchev, J. Yang, Construction of combustion models for rapeseed methyl ester bio-diesel fuel for internal combustion engine applications, *Biotechnol. Adv.* 27 (2009) 641–655.
- [15] H.K. Ng, S. Gan, J.-H. Ng, K.M. Pang, Development and validation of a reduced combined biodiesel-diesel reaction mechanism, *Fuel* 104 (2013) 620–634.
- [16] O. Herbinet, W.J. Pitz, C.K. Westbrook, Detailed chemical kinetic mechanism for the oxidation of biodiesel fuels blend surrogate, *Combust. Flame* 157 (2010) 893–908.
- [17] P. Gokulakrishnan, A.D. Lawrence, P.J. McLellan, E.W. Grandmaison, A functional-PCA approach for analyzing and reducing complex chemical mechanisms, *Comput. Chem. Eng.* 30 (2006) 1093–1101.
- [18] S. Vajda, P. Valko, T. Turányi, Principal component analysis of kinetic models, *Int. J. Chem. Kinet.* 17 (2010) 55–81.
- [19] Y. Liu, Y. Wu, Y. Gao, T. Lu, A linearized error propagation model based on Jacobian analysis for skeletal mechanism reduction, 9th U.S. National Combustion Meeting (2015) paper RK38.
- [20] Y. Zhang, M.A. Dubé, D.D. McLean, M. Kates, Biodiesel production from waste cooking oil: 2. Economic assessment and sensitivity analysis, *Bioresour. Technol.* 90 (2003) 229–240.
- [21] A.S. Tomlin, M.J. Pilling, T. Turányi, J.H. Merkin, J. Brindley, Mechanism reduction for the oscillatory oxidation of hydrogen: sensitivity and quasi-steady-state analyses, *Combust. Flame* 91 (1992) 107–130.
- [22] T. Lu, C.K. Law, A directed relation graph method for mechanism reduction, *Proc. Combust. Inst.* 30 (2005) 1333–1341.
- [23] T. Lu, C.K. Law, On the applicability of directed relation graphs to the reduction of reaction mechanisms, *Combust. Flame* 146 (2006) 472–483.
- [24] J. An, Y. Jiang, Differences between direct relation graph and error-propagation-based reduction methods for large hydrocarbons, *Proc. Eng.* 62 (2013) 342–349.
- [25] P. Pepiot-Desjardins, H. Pitsch, An efficient error-propagation-based reduction method for large chemical kinetic mechanisms, *Combust. Flame* 154 (2008) 67–81.
- [26] T. Lu, M. Plomer, Z. Luo, S.M. Sarathy, W.J. Pitz, Directed relation graph with expert knowledge for skeletal mechanism reduction, 7th U.S. National Combustion Meeting (2011) paper 1F01.
- [27] W. Sun, Z. Chen, X. Gou, Y. Ju, A path flux analysis method for the reduction of detailed chemical kinetic mechanisms, *Combust. Flame* 157 (2010) 1298–1307.
- [28] G. Li, H. Rabitz, J. Tóth, A general analysis of exact nonlinear lumping in chemical kinetics, *Chem. Eng. Sci.* 49 (1994) 343–361.
- [29] R. Fournet, V. Warth, P.A. Glaude, F. Battin-Leclerc, G. Scacchi, G.M. Côme, Automatic reduction of detailed mechanisms of combustion of alkanes by chemical lumping, *Int. J. Chem. Kinet.* 32 (2015) 36–51.
- [30] Z. Luo, T. Lu, M.J. Maciaszek, S. Som, D.E. Longman, A reduced mechanism for high-temperature oxidation of biodiesel surrogates, *Energy Fuel* 24 (2010) 6283–6293.
- [31] J.L. Brakora, Y. Ra, R.D. Reitz, J. McFarlane, C.S. Daw, Development and validation of a reduced reaction mechanism for biodiesel-fueled engine simulations, *SAE Int. J. Fuel Lubr.* 1 (2008) 675–702.
- [32] J.L. Brakora, R.D. Reitz, Investigation of NOx predictions from biodiesel fueled HCCI engine simulations using a reduced kinetic mechanism, *SAE Technical Paper*, 2010-01-0577, 2010.
- [33] J.L. Brakora, Y. Ra, R.D. Reitz, Combustion model for biodiesel-fueled engine simulations using realistic chemistry and physical properties, *SAE Int. J. Eng.* 4 (2011) 931–947.
- [34] Z. Luo, M. Plomer, T. Lu, S. Som, D.E. Longman, A reduced mechanism for biodiesel surrogates with low temperature chemistry for compression ignition engine applications, *Combust. Theory Model.* 16 (2012) 369–385.
- [35] Z. Luo, M. Plomer, T. Lu, S. Som, D.E. Longman, S.M. Sarathy, W.J. Pitz, A reduced mechanism for biodiesel surrogates for compression ignition engine applications, *Fuel* 99 (2012) 143–153.
- [36] P. Pepiot-Desjardins, H. Pitsch, An automatic chemical lumping method for the reduction of large chemical kinetic mechanisms, *Combust. Theory Model.* 12 (2008) 1809–1118.
- [37] T. Lu, C.K. Law, Strategies for mechanism reduction for large hydrocarbons: n-heptane, *Combust. Flame* 154 (2008) 153–163.
- [38] S.-C. Kong, Y. Sun, R.D. Reitz, Modeling diesel spray flame liftoff, sooting tendency, and NO_x emissions using detailed chemistry with phenomenological soot model, *J. Eng. Gas Turb. Power* 129 (2007) 245–251.
- [39] J.-G. Nerva, C.L. Genzale, S. Kook, J.M. García-Oliver, L.M. Pickett, Fundamental spray and combustion measurements of soy methyl-ester biodiesel, *Int. J. Eng. Res.* 14 (2013) 373–390.
- [40] S.-C. Kong, Y. Sun, R.D. Reitz, Modeling diesel spray flame liftoff, sooting tendency, and NO_x emissions using detailed chemistry with phenomenological soot model, *J. Eng. Gas Turb. Power* 128 (2006) 245–251.
- [41] M.A. Patterson, R.D. Reitz, Modeling the effects of fuel spray characteristics on diesel engine combustion and emissions, *SAE Technical Paper* 980131, 1998.
- [42] L. Zhang, S.-C. Kong, Vaporization modeling of petroleum–biofuel drops using a hybrid multi-component approach, *Combust. Flame* 157 (2010) 2165–2174.